Replication Project:

"Household education gaps and gender role attitudes"

Giani, Hope & Skorge (2022)

Applied Statistical Analysis II

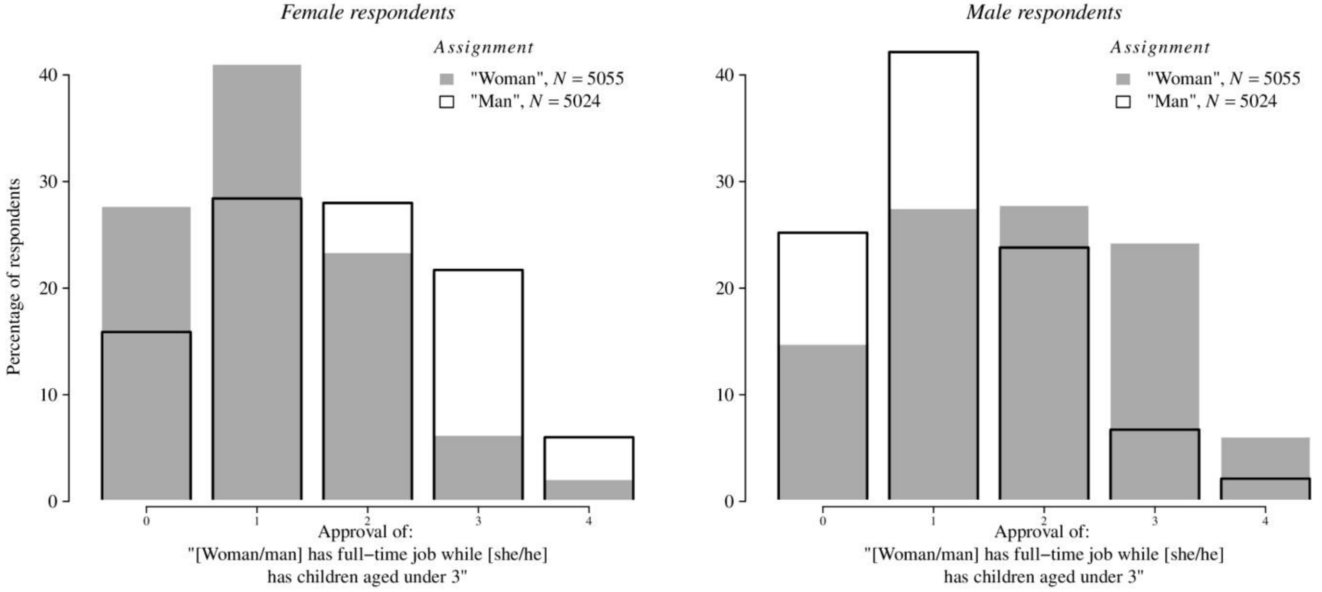
Presented by ***Matilda Tomatis***

Original Paper - Introduction

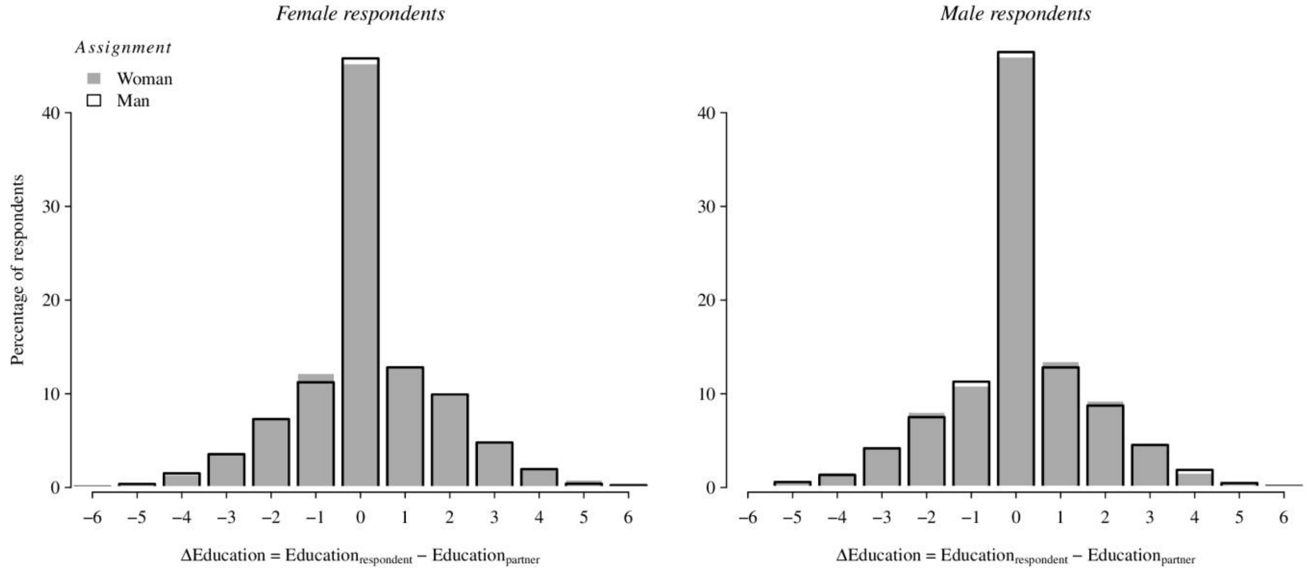
- It’s a 2022 paper, published in the *Political Science Research and Methods* by Giani, Hope & Skorge.
- It investigates how educational gap within the household mediate one’s gender childcare bias, hereby expressed as the difference in approval of a mother versus a father having a full-time job while having children under the age of three.
- The findings align with a resource-bargaining framework.

Data – European Social Survey

- 2018 ESS (Round 9) data, survey face-to-face carried out in 19 EU countries.
- <u>Dependent variable of interest</u>: *Approval*. On a scale from 0 (Strongly approve) to 4 (Strongly Disapprove).
- <u>Main independent variables</u>:
    - *Vignette Gender*: half of the respondents were asked about a mother and half about a father, binary variable.
    - *Household Education Gap:* the difference in the respondent education minus that of their partner, discrete variable spanning from -6 to +6.



Note: Approval ranges from Approval (1) to Disapproval (5).



Note: Positive values indicate respondents are more educated than their partner.

Original Model: Gaussian Kernel Interflex (Hainmueller, Mummolo and Xu, 2019)

The model allows the gender childcare bias to vary non-monotonically across levels of the household educational gap:

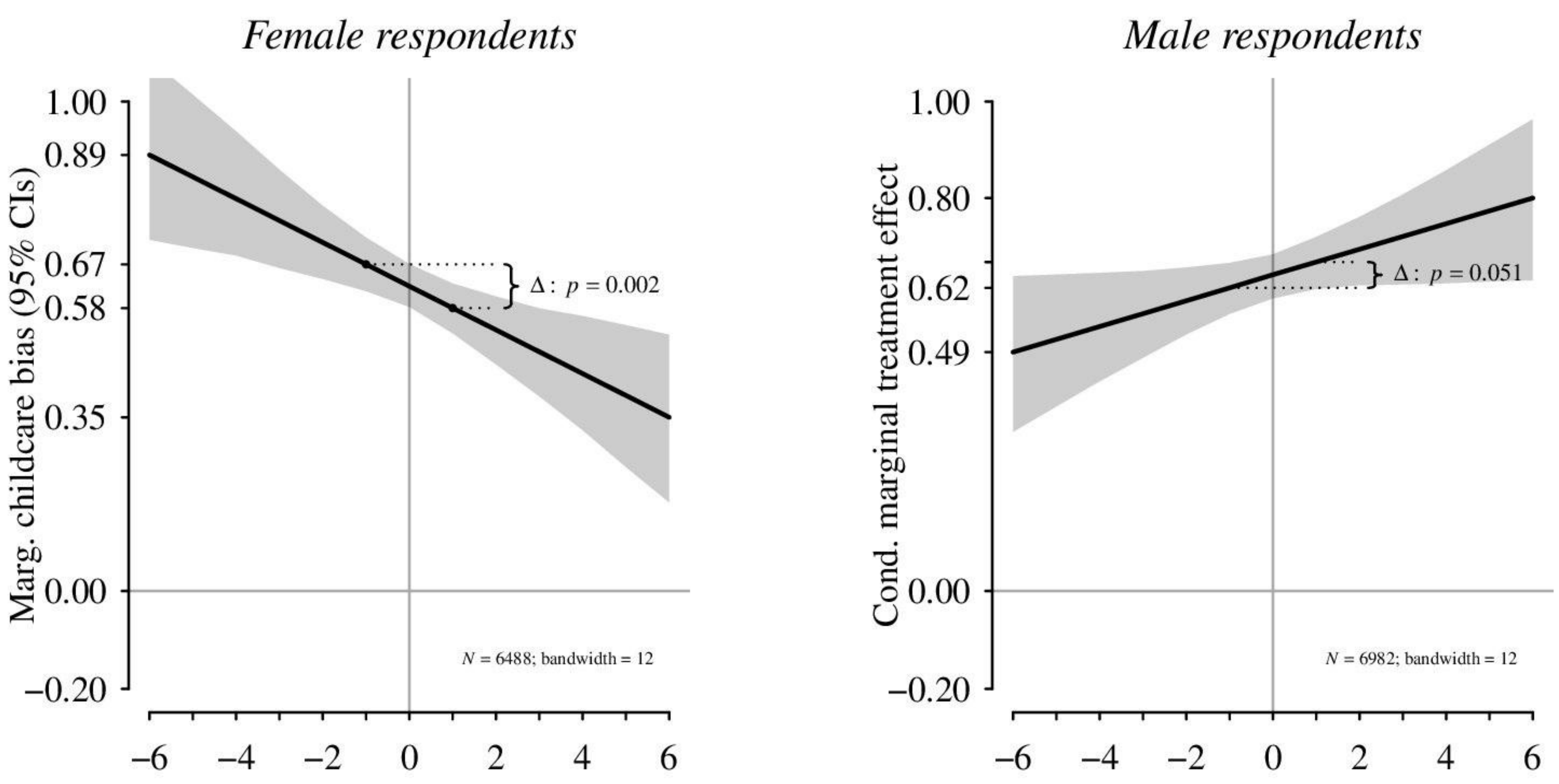
$$Y\{i,c\} \;=\; \alpha + \; f(\Delta E\{i,c\}) \;+\; g(\Delta E\{i,c\})\, B\{i,c\} \;+ \gamma(\Delta E\{i,c\})\, Z\{i,c\} \;+\; \mu c \;+\; \varepsilon i, c\}$$

The key quantity of interest is $g\,(\cdot)$: the marginal gender childcare bias, the difference in approval between a mother and a father holding a full-time job, conditional on $\Delta E_{i,c}$.

The sample is split by respondent gender to assess whether the pattern differs between men and women.

```
### Analysis with covariates for women --> core
ix.kern.z.w <- interflex(dt.w,
                         Y="Approval",
                         D="Vignette_gender",
                         X="Edu_diff",
                         Z=v.z,
                         estimator="kernel",
                         FE=c("Country"),
                         parallel=v.parallel,
                         cores=v.cores,
                         nboots=v.boots,
                         neval=13,
                         cutoffs = c(-6,-1,.9,6),
                         na.rm=TRUE,
                         wald=TRUE,
                         diff.values=c(-1,1),
                         bw = 12
)
## For men --> core
ix.kern.z.m <- interflex(dt.m,
                         Y="Approval",
                         D="Vignette_gender",
                         X="Edu_diff",
                         Z=v.z,
                         estimator="kernel",
                         FE=c("Country"),
                         parallel=v.parallel,
                         cores=v.cores,
                         nboots=v.boots,
                         neval=13,
                         cutoffs = c(-6,-1,.9,6),
                         na.rm=TRUE,
                         wald=TRUE,
                         diff.values=c(-1,1),
                         bw = 12

)
```



Main Findings

1. For *female respondents,* as the household education gap increase in their favour, their gender childcare bias tends to diminish.
2. For *male respondents* the opposite trend is observed: as the educational gap increases in their favour, they tend to have a bigger gender bias.
3. These findings are in line with a *resource-bargaining framework*: women who out-educate their partners hold more power in the household and hold more egalitarian attitudes; men show the opposite pattern.

My Contribution: Linear Interflex Model

```r
## Females --> Linear
ix.z.w.lin <- interflex(dt.w,
                        Y="Approval",
                        D="Vignette_gender",
                        X="Edu_diff",
                        FE=c("Country"),
                        Z=v.z,
                        estimator="linear",
                        parallel=v.parallel,
                        cores=v.cores,
                        nboots=v.boots,
                        diff.values=c(-1,1),
                        na.rm = TRUE
)

## Males --> Linear

ix.z.m.lin <- interflex(dt.m,
                        Y="Approval",
                        D="Vignette_gender",
                        X="Edu_diff",
                        Z=v.z,
                        estimator="linear",
                        FE=c("Country"),
                        parallel=v.parallel,
                        cores=v.cores,
                        nboots=v.boots,
                        diff.values=c(-1,1),
                        na.rm=TRUE,
)
```

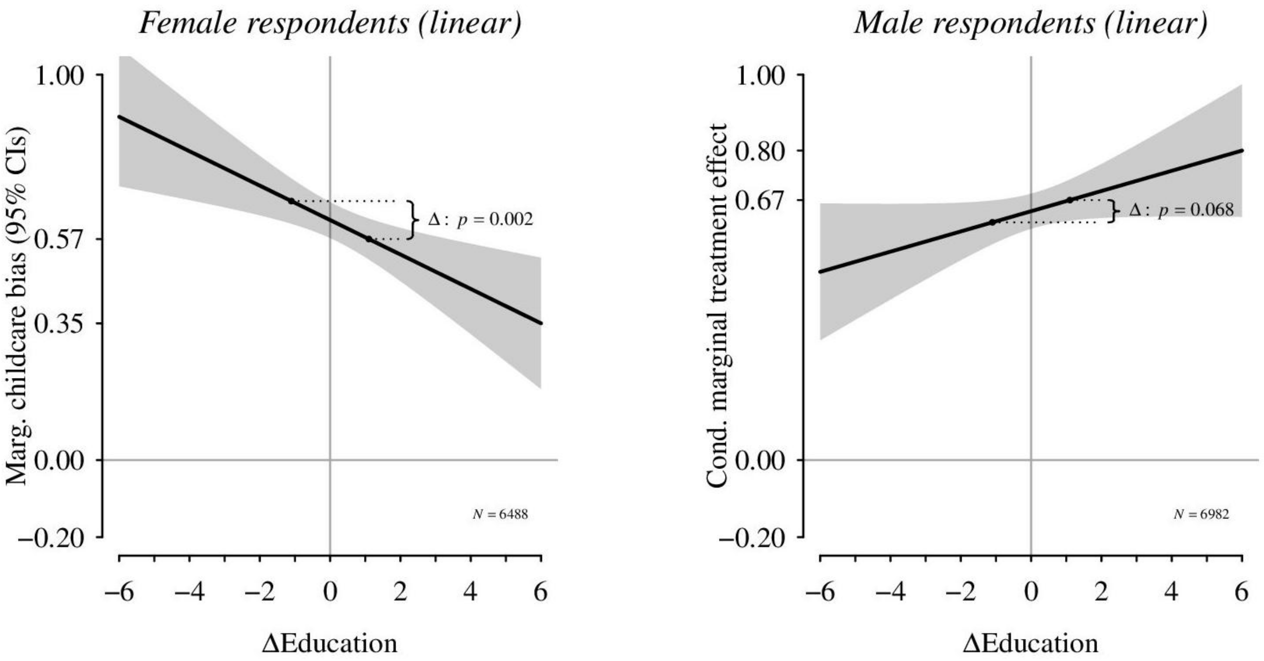
- The Gaussian kernel model is computationally intensive and its output, a smooth function $g(\cdot)$, is not straightforward to interpret.
- Given that the original results appeared linear, I fit a ***linear interflex model*** retaining the same covariates and country fixed effects, trading flexibility for parsimony and interpretability.

$$\boldsymbol{Y\{i,c\} = \alpha + \beta_1 B\{i,c\} + \beta_2 \Delta E\{i,c\} + \beta_3 (B\{i,c\}\Delta E\{i,c\}) + \gamma Z\{i,c\} + \mu c + \varepsilon\{i,c\}}$$

In this specification our focus is $\boldsymbol{\beta_3}$, the marginal gender childcare bias given a specific level of household educational gap.

Table 1: Linear Interflex Models - Core Explanatory Variables

	Dependent variable:	
	Approval	
	Women (Linear)	Men (Linear)
	(1)	(2)
Educational Gap (β_1)	0.035^{***}	0.013
	(0.011)	(0.011)
Vignette Gender (β_2)	0.623^{***}	0.646^{***}
	(0.024)	(0.023)
Edu. Gap × Vignette Gender (β_3)	-0.045^{***}	0.026^{*}
	(0.015)	(0.014)
Observations	6,488	6,982
R^2	0.258	0.245
Adjusted R^2	0.254	0.241
Residual Std. Error	0.943 (df = 6450)	0.961 (df = 6944)
Note:		$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$



My Findings

- The two models reports identical results for female respondents and strongly similar values for males
- Specifically, we find that has a $\boldsymbol{\beta_3}$ values of -0,045 for women and 0,026 for men; the base value, when the educational gap is equal to 0, is similar for both genders, validating the <u>*resource-bargaining*</u> framework .

<u>***In conclusion:***</u> The results are near-identical across both specifications, suggesting the added complexity of the kernel model was unnecessary.

Q & A

Thank you for your attention